Gaming Lower-Code Functionality

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Contents

1. Introduction

1.1. General Description:

"The project involves using a game engine called Godot to create a plug-in within the engine that automates designing interior levels, such as a room or rooms. The plug-in would allow the user to quickly build out basic levels that can be built off of rather than directly creating a comprehensive and highly detailed design. The plug-in would take in parameters that would then scaffold a basic game."

1.2. Product Scope:

This Godot plug-in will allow a user to select parameters from an on-screen form to develop a procedurally generated 3D room or set of rooms. Users may modify parameters between generations. Parameters include selections like adding furniture to populate a room, the number of rooms to generate, the distance between room walls, or the size of rooms. Once generated, users may modify their environments without necessitating further use of the plug-in. All room generations will be built using a simple snap grid layout.

Possible pitch for later development: Allow users to "lock" features of an already generated room such as the layout, placement of furniture, etc. This would allow users to generate a room, decide what they like, and modify other parameters as needed without discarding their selections.

Possible pitch for later development: Implement drag-and-drop movement of doors or furniture within a generated room.

Possible pitch for later development: Provide a predetermined structures rule set template. Examples: Linear room generation, a single long hallway with branches, rooms connecting to create a loop(s), etc. while still randomly generating the rooms and furnishings.

1.3. Intended Use:

The plug-in is intended to allow quick environment creation for users to adjust and populate at will. This will accelerate level creation and game development and ease the workload of developers.

1.4. Intended Audience:

All users of Godot software with the intent to develop 3D environments. Students who are learning to use Godot or game design.

1.5. Definitions

The following are the definitions of terminology used within the document:

TERMS	DEFINITION

2. Function Requirements

2.1. Performance Requirements:

- Allow users to make selections describing the rooms and levels they'd like to develop.
- Generate and regenerate rooms quickly and efficiently
- Create a snap grid for ease of use

2.2. Compliance Needs:

- Like Godot, this plug-in will be open-source and free to use
- No user data is collected

Possible pitch for later development: User manual/tutorial for accessibility purposes.

3. External Interfaces

3.1. User Interfaces:

A form on the screen collects user input for room generation. A “Generate” or “Regenerate” button at the bottom of the form allows the user to generate a room or regenerate a room with the same parameters for a different result.

3.2. Hardware Interfaces:

N/A.

3.3. Software Interfaces:

The software is to be built in GDScript as opposed to C#. It must be compatible with existing Godot nodes and interfaces.

3.4. Communication Interfaces:

Possible pitch for later development: Create a voluntary feedback option. Collect user data to analyze and address crashes, errors, or requests.

4. Non-functional Requirements

4.1. Security:

The plug-in does not access any user data.

4.2. Compatibility:

The plug-in will be compatible with existing Godot 4 engines.

Possible pitch for later development: Develop cross-compatibility with frequently used relevant plug-ins. For example, Godot users often use Aseprite to create graphics or sprites. Offer compatibility with Aseprite files when texturing rooms.

4.3. Availability:

The plug-in will be available 24/7 as it runs locally on the user's device and is hosted for download by Godot.

4.4. Scalability:

It shall be possible to provide updates for newer versions of the GODOT software.

4.5. Maintainability:

Maintain compatibility with new Godot engines as they are released.

4.6. Usability:

The plug-in will be suitable for beginning game designers in a classroom setting. A clean and intuitive interface will